Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/08/2007
PI Name:	Tosini, Gianluca Ph.D.		
Project Title:	Preventing Desynchronization of the Circadian System in Long-Term Space Flight		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Human Performance Factors, Sleep, and Chronobiology Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	30310-1495	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	09/01/2004	**End Date:**	08/31/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Fukuhara, Chiaki (Morehouse School of Medicine)		
Grant/Contract No.:	NCC 9-58-HPF00407		
Performance Goal No.:			
Performance Goal Text:			

Many biochemical, physiological and behavioral parameters exhibited by organisms show daily fluctuations and most of these daily rhythms persist in constant conditions, thus, demonstrating that they are driven by endogenous oscillators. The rhythms that persist in constant conditions with a period close to 24 hours are called circadian rhythms. One the most important aspects of space flight is the absence of geophysical 24-h cycles, which, of course, affects the overall temporal organization of the organisms. In the case of long-duration manned space flight, it is crucial to understand how the whole circadian system would react and behave in such circumstances.
We discovered that exposing rats to constant dim light for 60 days may induce spontaneous internal desynchronization in a few animals. In the present research proposal we further examined this phenomenon by investigating the physiological consequences of spontaneous internal desynchronization and to develop countermeasures to obviate the occurrence of internal desynchronization in animals exposed to constant dim light. Spontaneous internal

Task Description:	desynchronization has profound effects on the capability of the organisms to perform (mentally and physically) and to remain healthy. Recent studies have shown that a subset of retinal ganglion cells (RGCs) innervating the SCN are directly photosensitive and able to convert electromagnetic radiation into neural signals. Melanopsin, a photopigment based on vitamin A, was found in these RGCs and is the strongest candidate for the circadian photopigment within these cells. The spectral sensitivity of these RGC peak around 474 nm. It is now believed that these RGCs provide the main light input to the circadian clock. During the last year we have initiated a collaboration with Dr. Ian Fergusson (an expert in solid state lighting) at the Georgia Institute of Technology and we have developed a series of light blue narrow-band light-emitting diodes (LEDs) with an emission peak at 476 nm that should be very effective in stimulating the photosensitive retinal ganglion cells. Indeed, our preliminary data indicate that these LEDs may be a valid countermeasure to prevent disfunctions of the circadian system that may occur in the space flight. In the next year of funding we plan to further investigate the use of these LEDs.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our project will provide important data on the cellular, molecular and physiological mechanisms that are involved in the maintenance of the synchronization among the different circadian oscillators. The data that will be generated by this project will allow the design of specific treatments (pharmacological or environmental) that will prevent the desynchronization of the circadian system. Such treatments will then be employed in the treatment of circadian dysfunctions due to a pathological (or environmental) condition that may affect some individuals. Furthermore, the LEDs that we have developed may be useful in the treatment of several pathologies such as seasonal affective disorders and sleep disorders.
Task Progress:	The experiments that we have carried out during the second year of funding have provided important new data about the functional organization of the mammalian circadian system. As proposed in the original application, we have terminated the experiment described in Specific Aims 1 (to investigate the physiological consequences of pineal Aa-nat mRNA desynchronization) and 2 (to investigate the cellular and molecular mechanisms responsible for the desynchronization between locomotor activity pattern and melatonin), and we have already published some of these new results. We have obtained the brains that are needed to perform the experiments that focus on the cellular and molecular mechanisms responsible for the desynchronization between locomotor activity pattern and melatonin. Futhermore, we have developed a new series of blue narrow-band light-emitting diodes (LEDs) that are very effective in modulating some circadian parameters (melatonin suppression and phase-shift of the locomotor activity).
Bibliography Type:	Description: (Last Updated: 02/21/2008)
Articles in Peer-reviewed Journals	Aguzzi J, Bullock NM, Tosini G. "Spontaneous internal desynchronization of locomotor activity and body temperature rhythms from plasma melatonin rhythm in rats exposed to constant dim light." J Circadian Rhythms. 2006 Apr 4;4:6. PMID: 16594995 , Apr-2006
Articles in Peer-reviewed Journals	Tosini G, Aguzzi J. "Effect of space flight on circadian rhythms." Adv Space Biol Med. 2005;10:165-74. Review. PMID: 16101107 , Oct-2005
Articles in Peer-reviewed Journals	Sakamoto K, Liu C, Kasamatsu M, Iuvone PM, Tosini G. "Intraocular injection of kainic acid does not abolish the circadian rhythm of arylalkylamine N-acetyltransferase mRNA in rat photoreceptors." Mol Vis. 2006 Feb 23;12:117-24. PMID: 16518309 , Feb-2006
Articles in Peer-reviewed Journals	Sakamoto K, Liu C, Kasamatsu M, Pozdeyev NV, Iuvone PM, Tosini G. "Dopamine regulates melanopsin mRNA expression in intrinsically photosensitive retinal ganglion cells." Eur J Neurosci. 2005 Dec;22(12):3129-36. PMID: 16367779 , Dec-2005